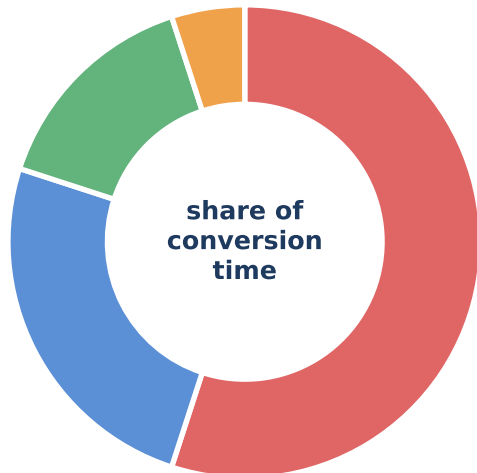


Conversion Time Breakdown by Pipeline Stage

profiled run, $n = 400$ rules, Ordering 4 — shares of wall-clock conversion time



55% Projection Normalization

Stage 5 · $O(N^2)$

25% Sweep-line cell construction

Stage 4, Phase 2 · $O(N \cdot n)$

15% Anomaly detection (Triad)

Stage 2 · $O(n^2)$

5% Expansion, action assignment, merge

Stage 4, Phases 1, 3, 4

N = cells before normalization; $N = O(n^3)$ per atomic protocol in the worst case (Section 4.4). Projection Normalization dominates at large n because of its $O(N^2)$ cost; conversion is a one-time offline cost. Absolute times for the released runs: Sections 6.4 and 6.9.